SMART INDIA HACKATHON 2026

TITLE PAGE

Problem Statement ID – 26104

Problem Statement Title– AI-Powered Real-Time Impersonation Attacks

Theme– Blockchain & Cybersecurity

PS Category– Software

Team ID– [placeholder]

Team Name (Registered on portal)– [placeholder]

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Problem Statement ID: **26104**

AI-Powered Real-Time Detection and Prevention of Voice Cloning Impersonation Attacks



VS



Real-Time Detection. Stronger Security. Safer Tomorrow.



PROPOSED SOLUTION

Our system detects AI-generated or cloned voices in real time to prevent impersonation and fraudulent activities.

VOICE INPUT



- Live phone call
- VoIP call
- Audio recording

PREPROCESSING

- Noise Reduction
- Normalization
- Segmentation
- Feature Extraction (MFCC, Spectrogram)

AI / ML MODEL ANALYSIS



Deep Learning Models (CNN, LSTM, ECAPA-TDNN)

Trained to detect AI / Cloned voices in real time

DECISION



System classifies the voice as

Authentic

or

Fake / Cloned

ACTION



- If Authentic: Allow Access
- If Fake / Cloned: Block Access & Generate Alert

AUTHENTIC VOICE



Access Granted

User Verified Successfully

FAKE / CLONED VOICE



Access Denied

Alert Generated

Security Team Notified

KEY FEATURES

- Real-Time Detection
- High Accuracy
- Fast Response
- Secure & Reliable



Prevents Impersonation & Fraud

Protects Sensitive Transactions

Ensures Trust & Digital Safety

TECHNICAL APPROACH

System Architecture

DATA SOURCES

- Live Phone Calls
- VoIP Calls
- Audio Recordings
- Audio Files

PREPROCESSING

- Noise Reduction
- Normalization
- Segmentation
- Feature Extraction (MFCC, Spectrogram)

AI MODEL ENGINE



Deep Learning Models
(CNN, LSTM, ECAPA-TDNN)

Trained to detect
AI / Cloned voices
in real time

DECISION & ACTION

AUTHENTIC VOICE
(Allow Access)

FAKE / CLONED VOICE
(Block Access & Generate Alert)

DATABASE

Store Voiceprints, Logs & Records

BLOCKCHAIN LEDGER

Tamper-proof logs for transparency & audit

ALERT SYSTEM

Real-time Alerts via SMS / Email / Dashboard

TECHNOLOGIES USED

Python | TensorFlow | Keras | OpenCV | LibROSA | MySQL | Blockchain (Hyperledger)

FEASIBILITY & VIABILITY

Our solution is technically feasible, economically viable and sustainable for real-world deployment.

TECHNICAL FEASIBILITY

- AI/ML models (CNN, LSTM, ECAPA-TDNN) already proven effective for voice spoofing detection.
- Real-time processing achievable with optimized models and cloud / edge infrastructure.
- High accuracy with continuous learning on diverse datasets.
- Integration with existing systems via APIs.

OPERATIONAL FEASIBILITY

- Easy integration with call centers, VoIP platforms and authentication systems.
- Real-time alerts enable immediate decision and action.
- User-friendly dashboard for monitoring and management.
- Minimal training required for operators.



ECONOMIC VIABILITY

- Reduces financial fraud and impersonation losses for organizations.
- Cost-effective compared to manual verification or post-incident investigation.
- Scalable solution suitable for banks, enterprises and government.
- Open-source tools and cloud services ensure cost optimization.

SUSTAINABILITY

- Model continuously improves with new data and feedback.
- Designed to adapt to new AI voice cloning techniques.
- Supports long-term security through regular updates and audit logs.
- Environmentally efficient with optimized computing and storage.

WHY IT IS VIABLE

- Addresses a critical and growing cybersecurity threat.
- High demand across multiple sectors.
- Scalable, reliable and future-ready solution.
- Strong potential for real-world impact and adoption.

Feasible Today. | Viable Tomorrow. | Secure Always.

Banks | Government | Enterprises | Individuals

IMPACT & BENEFITS

Our solution strengthens trust and security in voice-based interactions by detecting and preventing voice cloning impersonation attacks.

PREVENTS FINANCIAL FRAUD

Blocks unauthorized transactions, fake approvals and money scams caused by cloned voices.

PROTECTS BUSINESSES

Safeguards call centers, customer support, and enterprise systems from impersonation attacks.

REAL-TIME PROTECTION

Detects cloned voices instantly and prevents attacks before damage occurs.



ENHANCES TRUST & SAFETY

Builds confidence in voice-based communication for users and organizations.

SUPPORTS GOVERNANCE

Helps government and public sectors secure citizen services and sensitive communications.

WIDE APPLICATION

Useful in banking, telecom, healthcare, insurance, e-governance and any voice-enabled service.

KEY BENEFITS

- Fraud Prevention
- Data & Identity Security
- Real-Time Detection
- Operational Efficiency
- User Trust & Satisfaction
- Secure Digital Future

Securing Voices. Protecting Identities. Building a Trustworthy Digital World.

RESEARCH AND REFERENCES

AI-Powered Real-Time Detection and Prevention of Voice Cloning Impersonation Attacks

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TECHNOLOGY / RESOURCE	DESCRIPTION	SYSTEM USAGE	REFERENCE (WITH LINK)
ASVspoof Dataset	Dataset containing genuine and spoofed / deepfake speech samples.	Used to train and evaluate the model for detecting fake and cloned voices.	ASVspoof Challenge Official Website https://www.asvspoof.org/
VoxCeleb Dataset	Large-scale dataset with thousands of speakers from real-world videos.	Used for speaker verification and feature learning.	VoxCeleb Dataset (Oxford) https://www.robots.ox.ac.uk/~vgg/data/voxceleb/
LibriSpeech Dataset	Public speech corpus containing high-quality read speech recordings.	Used for training models and evaluating on clean speech data.	LibriSpeech https://www.openslr.org/12/
ASVspoof 2021 Database	Speech Deepfake and Anti-Spoofing Database released for research and evaluation.	Used for benchmarking and comparing model performance.	ASVspoof 2021 (Zenodo) https://zenodo.org/records/4835108
Deep Learning Models	Models like CNN, LSTM, ECAPA-TDNN can learn complex patterns in voice.	Used to classify audio as "Authentic" or "Fake / Cloned".	ASVspoof 2021 Baseline Systems (GitHub) https://github.com/asvspoof-challenge/2021
MFCC & Spectrogram	Feature extraction techniques that convert audio into meaningful representations.	Used in preprocessing to extract voice features for model input.	Librosa Documentation https://librosa.org/doc/latest/index.html
Scikit-learn	Machine learning library for classification, evaluation and performance metrics.	Used for traditional ML models, experiments and result analysis.	Scikit-learn Documentation https://scikit-learn.org/stable/
Research Articles & Papers	Recent studies on voice cloning, speech deepfakes and anti-spoofing techniques.	Used to understand attack methods, benchmarks and improve the proposed solution.	Google Scholar https://scholar.google.com/

REFERENCES DRIVE INNOVATION. RESEARCH BUILDS A SECURE FUTURE.

Thank You!